```
void operator()(ArgTypes... args);
```

Preconditions

`*this` has an associated task.

Effects

Invokes the associated task `func` as-if `INVOKE(func,args...)`. If the invocation returns normally, stores the return value in the asynchronous result associated with `*this`. If the invocation returns with an exception, stores the exception in the asynchronous result associated with `*this`.

Postconditions

The asynchronous result associated with `*this` is *ready* with a stored value or exception. Any threads blocked waiting for the asynchronous result are unblocked.

Throws

An exception of type `std::future_error` with an error code of `std::future_errc::promise_already_satisfied` if the asynchronous result already has a stored value or exception.

Synchronization

A successful call to the function call operator synchronizes-with a call to `std::future<ResultType>::get()` or `std::shared_future<ResultType>::get()`, which retrieves the value or exception stored.

Std::Packaged_Task::Make_Ready_At_Thread_Exit Member Function

Invokes the task associated with a `std::packaged_task` instance, and stores the return value or exception in the associated asynchronous result